Chapter 1

An Introduction to Macroeconomics

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Housekeeping

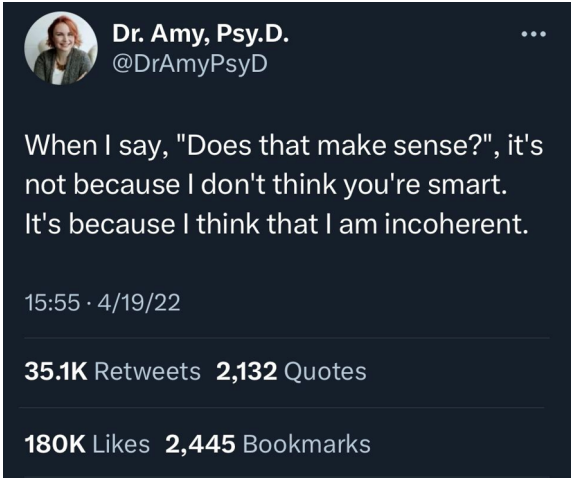
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Sorry to be annoying with this, but these did take a long time to make

Two points before we begin...



(Marschall, 2022) → When I say cite everything, I mean it!!!!

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What is Economics?

Economics is the study of the allocation of scarce resources and their production, distribution, and consumption

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What is Economics, Really?

Economics is *really* the study of human decision making and how those decisions influence resource allocation

What are these decisions?

- They are essentially microeconomics
- Modern macro is "micro-founded"
- It is built on microeconomics

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What Does This Mean for Macro?

Generally, we will study how micro-level decisions moves aggregate variables and prices

- National output does not make decisions; people do

We study how these decisions aggregate to influence things like total output, investment, consumption, etc.

- In the simplest form, aggregate means "to add up"
- *BUT*, everyone making a choice is different from one person making a choice $\rightarrow$ changes in prices vary

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Does This Matter?

If we want to study aggregate outcomes, why not just look at aggregates directly?

- Just look at total output before/after an event

We can, but this only provides us with a surface level view

- Can't see what's going on "under the hood"

Aggregates tell us *what*, not *why*

- Can't tell us what an untested policy will really do

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What do Microfoundations Tell Us?

People make decisions to maximize their expected utility

- Translation: people try their best, given any constraints, to make the choice they think will make them happiest

Possible constraints:

- Budget, time, space, energy, available information
  - E.g. smoking rates fell when more health information became available (Smith, 2008)

These decisions are also ***dynamic***; they occur over time

- Make choices today, based on expected budget tomorrow

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Dynamic Decisions: The Permanent Income Hypothesis

People don't spend out of their "current" budget, they spend out of their "Permanent" or "Lifetime" budget (Friedman, 1957)

- People prefer to "smooth" consumption over time
- Small hits to monthly lifetime budgets vs large hit to current budget
  - E.g. borrowing to go to college

Similar research on "Life-cycle" work (Modigliani and Brumberg, 1954)

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Research and Policy Implications

This has huge consequences for research and policy

If we want our models to accurately depict what we're trying to study, we need to be mindful of microfoundations

- Something to think about: is it enough for a model to make accurate predictions, or should it have accurate assumptions too? (Friedman, 1953)

If we want to conduct policy that helps people, we need to be mindful of how they will make decisions

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The Lucas Critique

What we know:

- People try to make the best choices given constraints
- These choices can change when the constraints change
- These decisions occur over time

If policy changes people's incentives, then we should expect decisions to change. Moreover, we should be aware of this when forming policy in the first place.

- Relationships estimated from historical data are confounded by historical policies

(Lucas, 1976)

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An Example with Taxes

A country is increasing consumption taxes to raise revenue

- People expect taxes to go up tomorrow
- Response: consume more today and less tomorrow to avoid higher costs (Cashin and Unayama, 2016)

With less consumption, there is less tax revenue

- Policy can respond by raising taxes even more to make up the difference...

But individuals' decisions will respond again

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Analysis Without Dynamic Responses

What would happen if we didn't consider the response to this policy?

If previous data says a 1pp increase in taxes leads to a 4% increase in revenue, should a 2pp increase in taxes lead to an 8% increase?

- Why or why not?
- We're looking at percent change, not the level of taxes. Does percent change account for the dynamics we're talking about?

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A Strict Example: Ricardian Equivalence

Consider a government trying to boost the economy through spending. They can pay for the spending by:

- Raising income taxes now
- Borrowing now and raising income taxes later

If people expect a fall in income, they will spend less now to save for the taxes coming in the future

Less household spending offsets government spending. The case where they cancel out is Ricardian Equivalence (Ricardo, 1820; Barro, 1974)

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Microfoundations Matter

Is Ricardian Equivalence strictly true? Probably not

- Ricardo didn't even think so. What most macroeconomists argue is the importance of being consistent with micro-behavior

Macro model building will have some type of optimization behavior for individuals in different settings (competitive, monopolistic, oligopolistic, etc.)

Then aggregate results and compare to real-world data

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The [quick] History of Macroeconomics

In an ironically recursive way, we can look at the study of macroeconomics as a series of dynamic decisions

We update our forecast of the field, which updates our forecasts of the macroeconomy, which updates our forecasts of the field...

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How We Got Here: Classical and Neoclassical Economics

1776: Adam Smith - "The Invisible Hand" (Smith, 1776)

1817: David Ricardo - comparative advantage vs mercantilism (Ricardo, 1817)

1870s: Carl Menger - marginalism and subjective theory of value (Menger, 1871)

- See also Walras, Marshall, Jevons (economics is "mathematical") (Jevons, 1862; Marshall, 1890)

1874: Léon Walras - General Equilibrium (Walras, 1874)

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How We Got Here: Keynesian Revolution

1936: John M. Keynes - "The General Theory" (Keynes, 1936)

- Written during Great Depression

Demand is a major driver of the economy

- Prices are **sticky**; they take time to adjust
- Wage stickiness prolongs unemployment
- Amplifies effects of poor demand

Government intervention can help:

- Raise taxes/cut spending in good times to build funds
- Lower taxes/raise spending in bad times to help unemployment

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How We Got Here: Neoclassical Synthesis

WWII - 1970: Consensus that in the short run, Keynesian analysis was correct. In the long run, markets equilibrate

- John Hicks and Alvin Hansen: IS-LM (Hicks, 1937; Hansen, 1949)
- Paul A. Samuelson: Modern basis for mathematical economics (Samuelson, 1947)
- Robert Solow: Long-run growth (Solow, 1956)

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How We Got Here: Monetarism

1960 - 1990: "Inflation is always and everywhere a monetary phenomenon"

- Milton Friedman and Anna Schwartz (Friedman and Schwartz, 1963; Friedman, 1970)

Revival of the Quantity Theory of Money and impact on inflation, as well as expectations of inflation

- Something Keynes thought of as second-order
- Very popular during high inflation episodes of the 1970s

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Today: Real Business Cycle Theory

1976 - today: Focus on real shocks causing changes to **relative prices** in equilibrium

- Price of one good in terms of another
- Assumes people view trade-offs in that sense
- Rational expectations about future values

Began with Lucas Critique; driven by Kydland and Prescott (Kydland and Prescott, 1982; Prescott, 1986; King et al., 1988)

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Today: New Keynesian Economics

1970s - today: Merging expectations with price stickiness (Fischer, 1977; Rotemberg, 1982; Calvo, 1983)

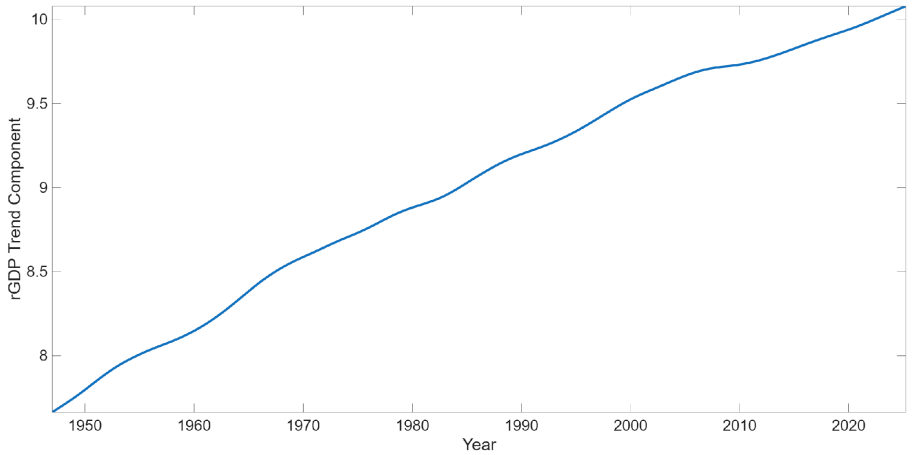
Elements of imperfect competition and coordination failures of the market (Diamond, 1982)

New Neoclassical Synthesis: Real Business Cycle Theory + New Keynesian Economics (Goodfriend and King, 1997)

- What we'll build to this semester

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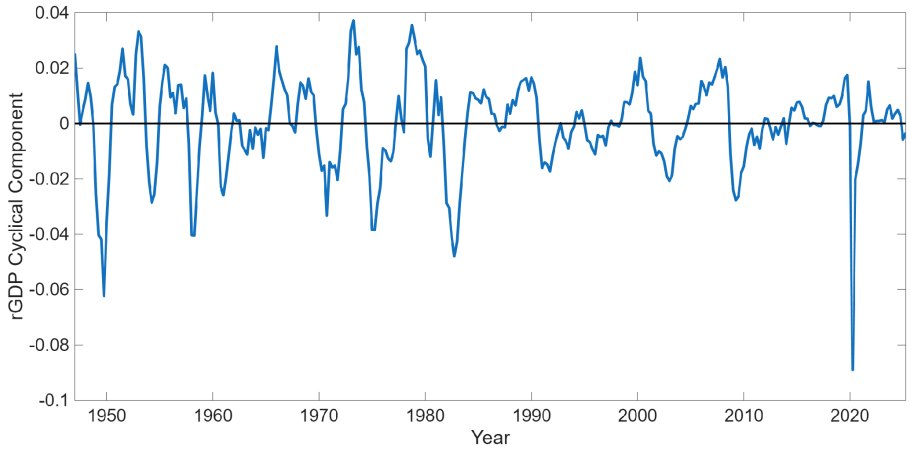
Today: In the Long Run, Markets Trend Up,...



HP filtered data of log of quarterly U.S. real GDP (chained 2017 dollars, seasonally adjusted annual rate); $\lambda = 1600$. Data from (BEA, 2026)

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... but in the Short Run, Imperfections Matter



HP filtered data of log of quarterly U.S. real GDP (chained 2017 dollars, seasonally adjusted annual rate); $\lambda = 1600$. Data from (BEA, 2026)

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Alternative Approaches (not covered)

Socialist economics:

- Abba Lerner
- Oskar Lange

Laissez-faire:

- Friedrich Hayek
- Ludwig von Mises

Debates about Economic Calculation: (Lange, 1936; Hayek, 1945)

- Do markets accurately tell us how people value resources?
- If not, what can?

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Beyond the Stock Market and Production

Four decades of topics not traditionally thought of as Economics:

Reparations and Racial Wealth Dynasties: Job Boerma and Loukas Karabarbounis. Reparations and persistent racial wealth gaps. *NBER Macroeconomics Annual*, 37(1):171–221, 2023

Traffic congestion's impact on birth weights *(fan favorite)*: Janet Currie and Reed Walker. Traffic congestion and infant health: Evidence from E-ZPass. *American Economic Journal: Applied Economics*, 3(1):65–90, 2011

Moneyball, viewed through economics: Jahn K. Hakes and Raymond D. Sauer. An economic evaluation of the Moneyball hypothesis. *Journal of Economic Perspectives*, 20(3):173–185, 2006

Experiments on how people perceive fairness under uncertainty: John H. Kagel, Chung Kim, and Donald Moser. Fairness in ultimatum games with asymmetric information and asymmetric payoffs. *Games and Economic Behavior*, 13(1):100–110, 1996

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